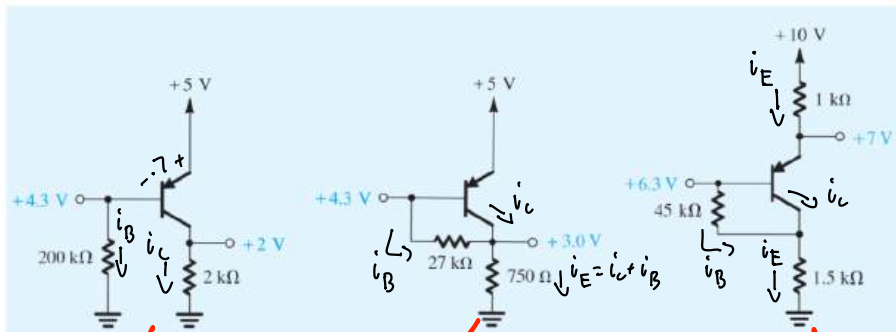


1. Measurements on the circuits below produce labeled voltages as indicated. Find the value of β for each transistor.



$$i_B = \frac{i_C}{\beta}$$

$$\beta = \frac{i_C}{i_B} = \frac{\left(\frac{2}{2k}\right)}{\left(\frac{4.3}{200k}\right)}$$

$$\beta = 414.512$$

$$i_B = \frac{i_E}{\beta + 1}$$

$$\beta = \frac{i_E}{i_B} - 1$$

$$\beta = \frac{\left(\frac{3}{750}\right)}{\left(\frac{4.3-3}{27k}\right)} - 1$$

$$\beta = 82.08$$

$$i_E = \frac{10-7}{1k} = 3mA$$

$$6.3 = i_B 45k + i_E 1.5k$$

$$\frac{6.3 - i_E 1.5k}{45k} = i_B$$

$$i_B = .04mA$$

$$i_C = i_E - i_B$$

$$= 3 - .04$$

$$i_C = 2.96mA$$

$$\beta = \frac{i_C}{i_B} = \frac{2.96}{.04} = 74$$

2. For the circuits in below, find values for the labeled node voltages and branch currents. Assume $\beta=100$ to be very high.

$$i_B = \frac{i_E}{\beta + 1} = \frac{.5mA}{100+1}$$

$$i_B \approx 4.95\mu A$$

$$i_C = \beta i_B \approx .495mA$$

$$\frac{3-V_2}{3.6k} = i_C$$

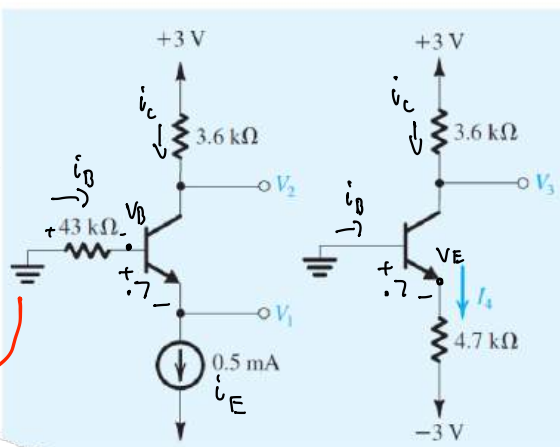
$$3 - i_C 3.6k = V_2$$

$$V_2 \approx 1.218[V]$$

$$V_D = -i_B 43k$$

$$V_D = -0.213$$

$$V_1 = V_D - .7 \approx -.413[V]$$



$$V_E = -0.7$$

$$I_4 = \frac{-0.7 - (-3)}{4.7k} = .489mA$$

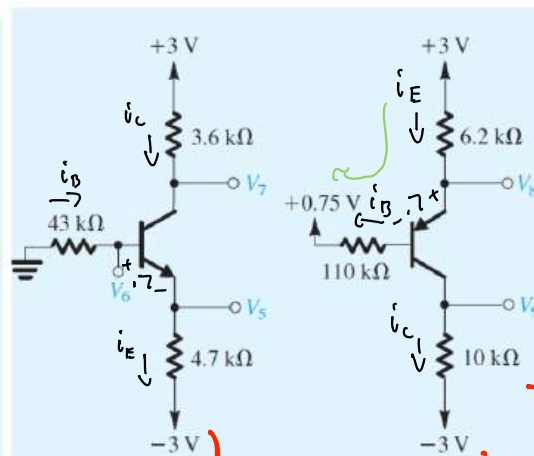
$$i_B = \frac{I_4}{\beta + 1} = 4.85\mu A$$

$$i_C = \beta i_B = .485mA$$

$$3 - V_7 = i_C 3.6k$$

$$V_7 = 3 - (.485mA)(3.6k)$$

$$V_7 \approx 1.26V$$



$$0 - (-3) = i_B 43k + .7 + i_E (4.7k)$$

$$3 - .7 = i_B (43k + (\beta + 1)(4.7k))$$

$$i_B = \frac{2.3}{(43k + (\beta + 1)(4.7k))}$$

$$i_B \approx 4.44\mu A$$

$$i_C = \beta i_B \approx 0.444mA$$

$$0 - V_4 = i_B 43k$$

$$V_4 = -0.191[V]$$

$$V_4 - .7 = V_5$$

$$V_5 = -0.891[V]$$

$$3 - V_7 = i_C 3.6k$$

$$V_7 = 3 - i_C 3.6k$$

$$V_7 \approx 1.4[V]$$

$$3 - .75 = i_E 6.2k + .7 + i_B 110k$$

$$2.25 - .7 = i_B (110k + (\beta + 1)6.2k)$$

$$i_B = \frac{1.55}{(110k + (\beta + 1)6.2k)}$$

$$i_B \approx 2.11\mu A$$

$$i_E = (\beta + 1) i_B = 0.213mA$$

$$3 - V_8 = i_E 6.2k \Rightarrow 3 - i_E 6.2k$$

$$V_8 \approx 1.682[V]$$

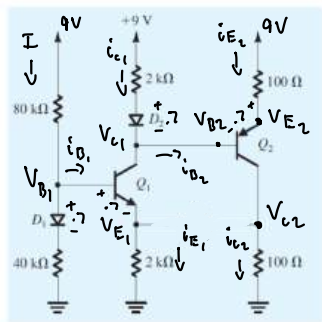
$$i_C = \beta i_B \approx .211mA$$

$$V_9 - (-3) = i_C 10k$$

$$V_9 = i_C 10k - 3 \approx -0.895[V]$$

3. Consider the circuit shown in figure below. Using $|V_{BE}|$ and $V_D = 0.7$ V independent of current, and $\beta = \infty$, find the voltages V_{B1} , V_{E1} , V_{C1} , V_{B2} , V_{E2} , and V_{C2} , initially with R open-circuited and then with R connected. Repeat for $\beta = 100$, with R open-circuited initially, then connected.

$$\beta = \infty$$



$$\frac{9-0.7}{120k} = I \approx 69.2 \mu A$$

$$9 - V_{B1} = I \cdot 80k$$

$$V_{B1} = 9 - 80k \cdot I$$

$$V_{B1} \approx 3.467 \text{ V}$$

$$V_{E1} = V_{B1} - 0.7 \approx 2.767 \text{ V}$$

$$i_{B1} = \frac{i_{C1}}{\beta} = \frac{i_{C1}}{\infty} = 0 \Rightarrow i_{C1} = i_{E1} \Rightarrow i_{E2} = i_{C2}$$

$$V_{E1} = i_{E1} \cdot 2k$$

$$i_{E1} = \frac{V_{E1}}{2k} \Rightarrow i_{E1} = i_{C1} = 1.38 \text{ mA}$$

$$9 - 0.7 - V_{C1} = i_{C1} \cdot 2k$$

$$8.3 - i_{C1} \cdot 2k = V_{C1}$$

$$V_{C1} \approx 5.533 \text{ V} = V_{B2}$$

$$V_{E2} - 0.7 = V_{B2}$$

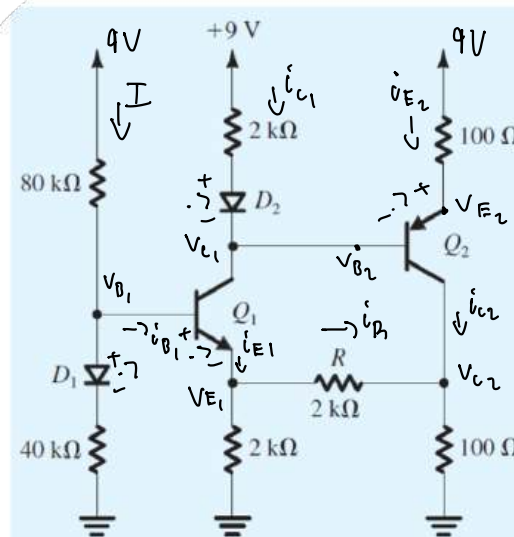
$$V_{E2} = V_{B2} + 0.7 = 6.23 \text{ V}$$

$$i_{E2} = \frac{9 - V_{E2}}{100} = 27.67 \mu A = i_{C2}$$

$$V_{C2} = i_{C2} \cdot 100$$

$$V_{C2} = 2.767 \text{ V}$$

$$\beta = \infty$$



$$V_{E1} = 2.767 \text{ V}$$

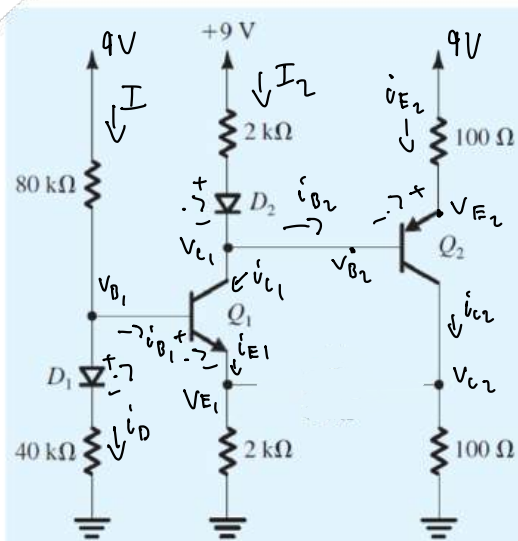
$$V_{C2} = 2.767 \text{ V}$$

$$i_{B1} = \frac{V_{E1} - V_{C2}}{R}$$

$$i_{B1} = 0$$

Voltages same whether R is connected or not ($\beta \rightarrow \infty$)

$$\beta = 100$$



$$V_{E2} = V_{B2} + 0.7 = 5.982 \text{ V}$$

$$i_{E2} = \frac{9 - V_{E2}}{100} = 30.18 \mu A$$

$$i_{C2} = \frac{\beta}{\beta + 1} i_{E2} = 29.88 \mu A$$

$$V_{C2} = i_{C2} \cdot 100 = 2.98 \text{ V}$$

$$V_{E1} = 2k(i_{B1} (101)) = 2.444 \text{ V}$$

$$KCL @ V_{B1}: I = i_{B1} + i_D$$

$$i_{E1} = i_B (\beta + 1) = \frac{V_{E1}}{2k} = \frac{V_{B1} - 0.7}{2k}$$

$$\frac{9 - V_{B1}}{80k} = \frac{V_{B1} - 0.7}{2k(101)} + \frac{V_{B1} - 0.7}{40k}$$

$$\frac{9}{80k} + \frac{0.7}{2k(101)} + \frac{0.7}{40k} = V_{B1} \left(\frac{1}{80k} + \frac{1}{2k(101)} + \frac{1}{40k} \right)$$

$$V_{B1} = \left(\frac{9}{80k} + \frac{0.7}{2k(101)} + \frac{0.7}{40k} \right) \left(\frac{1}{80k} + \frac{1}{2k(101)} + \frac{1}{40k} \right)^{-1}$$

$$V_{B1} = 3.144 \text{ V}$$

$$i_{B1} = \frac{V_{B1} - 0.7}{2k(101)} = 12.1 \mu A \Rightarrow i_{C1} = 1.21 \text{ mA}$$

$$KCL @ V_{C1}: I_2 = i_{C1} + i_{B2} \quad i_{B2} = \frac{i_{E2}}{\beta + 1} = \frac{9 - 0.7 - V_{C1}}{100(101)}$$

$$\frac{9 - 0.7 - V_{C1}}{2k} = i_{C1} + i_{B2}$$

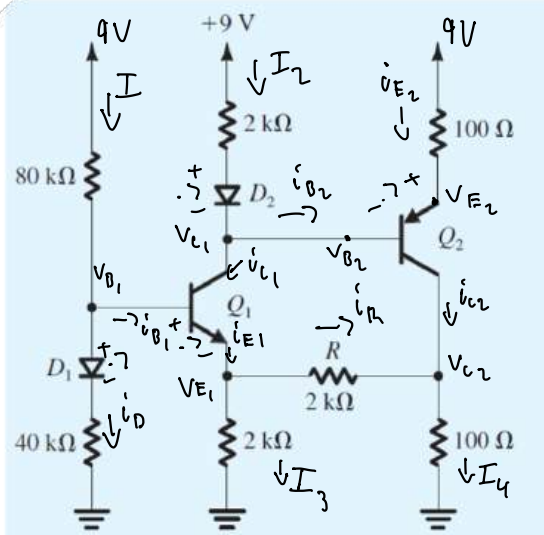
$$i_{C1} = \frac{8.3 - V_{C1}}{2k} - \frac{8.3 - V_{C1}}{100(101)}$$

$$i_{C1} - \frac{8.3}{2k} + \frac{8.3}{100(101)} = V_{C1} \left(-\frac{1}{2k} + \frac{1}{100(101)} \right) \Rightarrow$$

$$V_{C1} = \left(1.21 \text{ mA} - \frac{8.3}{2k} + \frac{8.3}{100(101)} \right) \left(-\frac{1}{2k} + \frac{1}{100(101)} \right)^{-1}$$

$$V_{C1} = 5.282 \text{ V} = V_{B2}$$

$$\beta = 100$$



$$KCL \text{ at } V_{c1}: \frac{9 - 7 - V_{c1}}{2k} = i_{c1} + i_{B2}$$

$$i_{c1} = \frac{\beta}{\beta + 1} i_{E1}$$

$$= \frac{100}{101} \left(\frac{V_{B1} - 7}{2k} + \frac{(V_{B1} - 7) - V_{c2}}{2k} \right)$$

$$\frac{V_{c1} - 8.3}{2k} + \frac{100}{101} \left(\frac{V_{B1} - 7}{2k} + \frac{(V_{B1} - 7) - V_{c2}}{2k} \right) + \frac{8.3 - V_{c1}}{100(101)} = 0$$

$$i_{B2} = \frac{i_{E2}}{\beta + 1} = \frac{9 - 7 - V_{c1}}{100(101)}$$

$$V_{c1} \left(\frac{1}{2k} - \frac{1}{100(101)} \right) + V_{B1} \left(\frac{100}{2k(101)} + \frac{100}{2k(101)} \right) - V_{c2} \left(\frac{100}{2k(101)} \right) = \frac{1.4 \cdot 100}{2k(101)} - \frac{8.3}{100(101)} + \frac{8.3}{2k}$$

$$V_{c1}(4000.99 \mu) + V_{B1}(490.099 \mu) - V_{c2}(495.05 \mu) = 4 \mu$$

$$KCL \text{ at } V_{c2}: \frac{V_{c2}}{100} = \frac{V_{E1} - V_{c2}}{2k} + i_{c2}$$

$$i_{c2} = \frac{\beta}{\beta + 1} i_{E2} = \frac{8.3 - V_{c1}}{101}$$

$$\frac{V_{c2}}{100} + \frac{V_{c2} - (V_{B1} - 7)}{2k} + \frac{V_{c1} - 8.3}{101} = 0$$

$$V_{c2} \left(\frac{1}{100} + \frac{1}{2k} \right) - V_{B1} \left(\frac{1}{2k} \right) + V_{c1} \left(\frac{1}{101} \right) = \frac{8.3}{101} - \frac{7}{2k}$$

$$-.5mV_{B1} + 9.9mV_{c1} + 10.5mV_{c2} = 81.8m$$

$$KCL \text{ at } V_{B1}: \frac{9 - V_{B1}}{80k} = \frac{V_{B1} - 7}{40k} + i_{B1}$$

$$i_{B1} = \frac{i_{E1}}{\beta + 1} = \left(\frac{V_{E1}}{2k} + \frac{V_{E1} - V_{c2}}{2k} \right) \frac{1}{\beta + 1} = \frac{1}{101} \left(\frac{V_{B1} - 7}{2k} + \frac{(V_{B1} - 7) - V_{c2}}{2k} \right)$$

$$\frac{V_{B1} - 9}{80k} + \frac{V_{B1} - 7}{40k} + \frac{1}{101} \left(\frac{V_{B1} - 7}{2k} + \frac{V_{B1} - 7 - V_{c2}}{2k} \right) = 0$$

$$V_{B1} \left(\frac{1}{80k} + \frac{1}{40k} + \frac{1}{2k(101)} + \frac{1}{2k(101)} \right) - V_{c2} \left(\frac{1}{2k(101)} \right) = \frac{9}{80k} + \frac{7}{40k} + \frac{7}{2k(101)} + \frac{7}{2k(101)}$$

$$V_{B1}(47.401 \mu) - V_{c2}(4.95 \mu) = 134.931 \mu$$

$$\begin{bmatrix} 47.401 \mu & 0 & -4.95 \mu \\ 990.099 \mu & 400.99 \mu & -495.05 \mu \\ -.5m & 9.9m & 10.5m \end{bmatrix}^{-1} \begin{bmatrix} 134.931 \mu \\ 4 \mu \\ 81.8m \end{bmatrix} = \begin{bmatrix} V_{B1} \\ V_{c1} \\ V_{c2} \end{bmatrix} \Rightarrow$$

$$V_{B1} \approx 3.17 \text{ V}$$

$$V_{c1} \approx 5.55 \text{ V} = V_{B2}$$

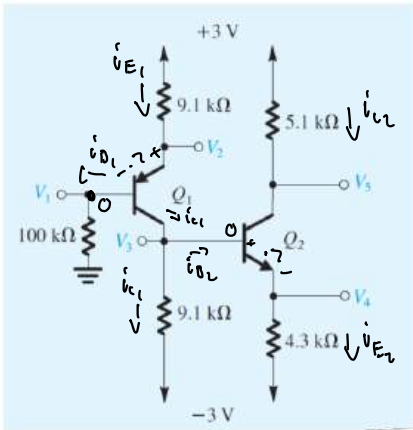
$$V_{c2} \approx 2.71 \text{ V}$$

$$V_{E1} = V_{B1} - 7 \approx 2.47 \text{ V}$$

$$V_{E2} = V_{B2} + 7 = 6.25 \text{ V}$$

4. For the circuit shown below, find the labeled node voltages for:

a. $\beta = \infty$
 $i_B = 0, \quad i_C = i_E$



$$V_1 = 0 \text{ [V]}$$

$$V_2 = 0.7 \text{ [V]}$$

$$\frac{V_3 - (-3)}{4.1\text{k}} = i_{C1} = i_{E1} = \frac{3 - 0.7}{4.1\text{k}}$$

$$V_3 + 3 = 2.3$$

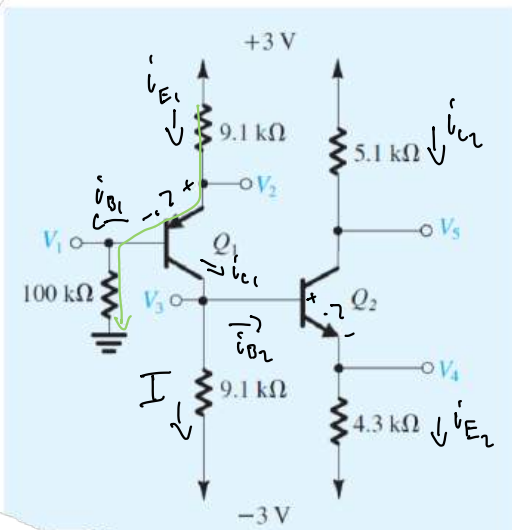
$$V_3 = -0.7 \text{ [V]}$$

$$V_4 = V_3 - 0.7 = -1.4 \text{ [V]}$$

$$\frac{3 - V_5}{5.1\text{k}} = i_{C2} = i_{E2} = \frac{V_4 + 3}{4.3\text{k}}$$

$$V_5 = 3 - \left(\frac{5.1}{4.3}\right)(-1.4 + 3) = 1.1 \text{ [V]}$$

b. $\beta = 100$



$$\text{KVL: } 3 - 0 = i_{E1} \cdot 9.1\text{k} + 0.7 + i_{B1} \cdot 100\text{k}$$

$$2.3 = i_{B1} \cdot ((\beta + 1) \cdot 9.1\text{k} + 100\text{k})$$

$$i_{B1} = \frac{2.3}{(101) \cdot 9.1\text{k} + 100\text{k}} = 2.257 \mu\text{A} \Rightarrow i_{E1} = i_B (\beta + 1) = 0.228 \text{ mA}$$

$$i_{C1} = \beta i_{B1} = 0.2257 \text{ mA}$$

$$3 - V_2 = i_{E1} \cdot 9.1\text{k}$$

$$V_2 = 3 - 2.07 = 0.926 \text{ [V]}$$

$$V_1 = V_2 - 0.7 = 0.226 \text{ [V]}$$

$$i_{C1} = I + i_{B2} \quad i_{B2} = \frac{i_{E2}}{\beta + 1}$$

$$i_{C1} = \frac{V_3 - (-3)}{9.1\text{k}} + \frac{V_3 + 2.3}{4.3\text{k}(101)} = \frac{V_3 - 0.7 - (-3)}{4.3\text{k}(101)}$$

$$i_{C1} = V_3 \left(\frac{1}{9.1\text{k}} + \frac{1}{4.3\text{k}(101)} \right) + \frac{3}{9.1\text{k}} + \frac{2.3}{4.3\text{k}(101)}$$

$$V_3 = \frac{\left(i_{C1} - \frac{3}{9.1\text{k}} - \frac{2.3}{4.3\text{k}(101)} \right)}{\frac{1}{9.1\text{k}} + \frac{1}{4.3\text{k}(101)}} = -0.974 \text{ [V]}$$

$$3 - V_5 = i_{C2} \cdot 5.1\text{k}$$

$$V_5 = 3 - \frac{\beta}{\beta + 1} \left(\frac{V_4 - (-3)}{4.3\text{k}} \right) \cdot 5.1\text{k}$$

$$V_5 = 1.443 \text{ [V]}$$

5. Given the expression

$$H(j\omega) = \frac{z_1 z_2}{p_1 p_2 p_3} = \frac{-\left(\frac{\omega}{100}\right)^2}{\left(1 + j\frac{\omega}{100}\right)\left(1 + j\frac{\omega}{1,000}\right)\left(1 + j\frac{\omega}{10,000}\right)}$$

$$j^2\left(\frac{\omega}{100}\right)^2 = -\left(\frac{\omega}{100}\right)^2$$

- Find the starting slope of the Bode plot,
- Find the mid-band or pass-band gain,
- Determine the total phase shift,
- Plot the straight-line approximation for the magnitude and phase, and
- Plot the complete transfer function using your favorite software (phase and magnitude).
- What happened to the j in the numerator?

a) $|H(j\omega)| \approx \left(\frac{1}{10}\right)^2$

$$20 \log\left(\frac{1}{10}\right)^2 = 40 \log(1) - 40 \log(10)$$

$$40 \text{ dB/dec}$$

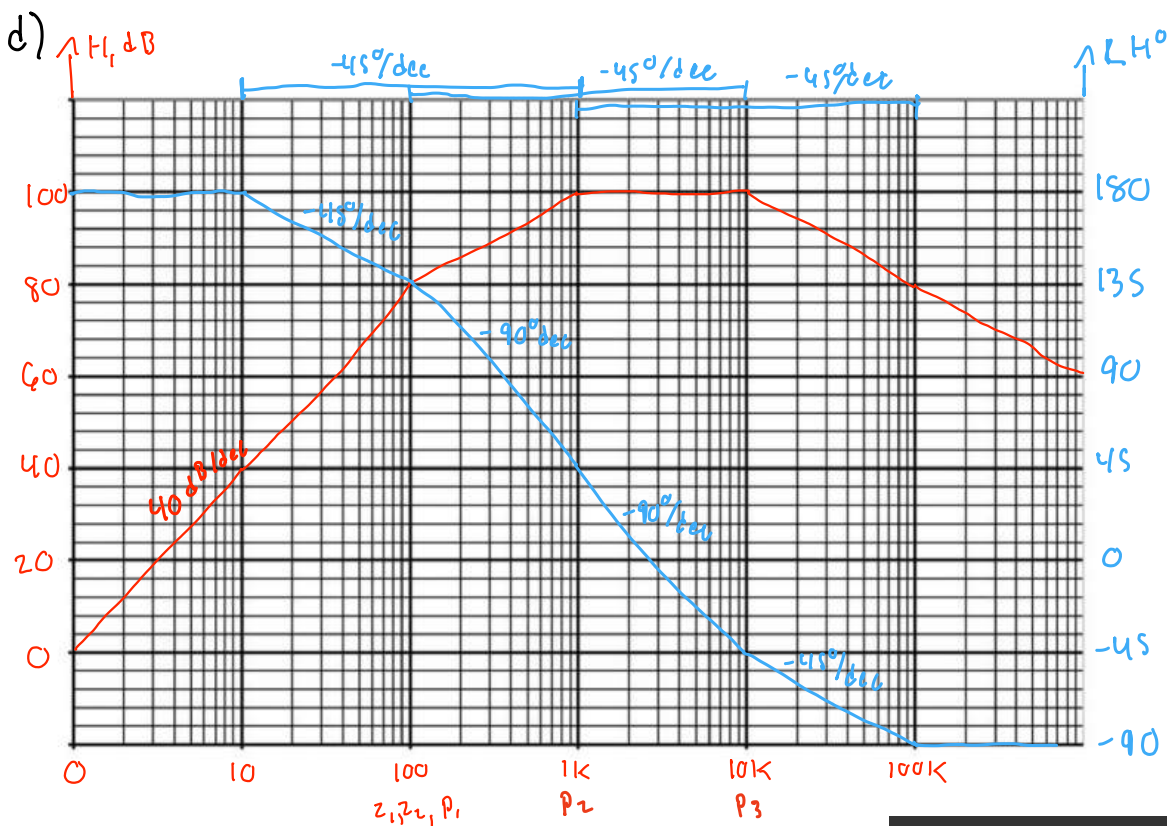
b) $1K < \omega < 10K$

$$H \approx \frac{-\left(\frac{\omega}{100}\right)^2}{\left(\frac{j\omega}{100}\right)\left(\frac{j\omega}{1000}\right)(1)} = \frac{-\frac{\omega^2}{10K}}{-\frac{\omega^2}{100K}} = \frac{100K}{10K} = 10$$

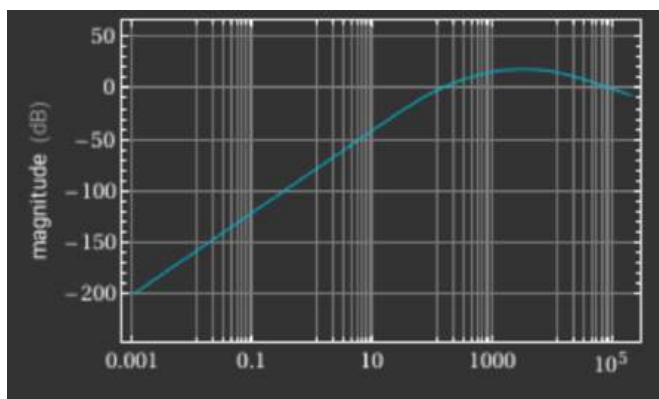
$$20 \log(10) = 20 \text{ dB}$$

c) $+180 - 90 - 90 - 90 = -90^\circ$

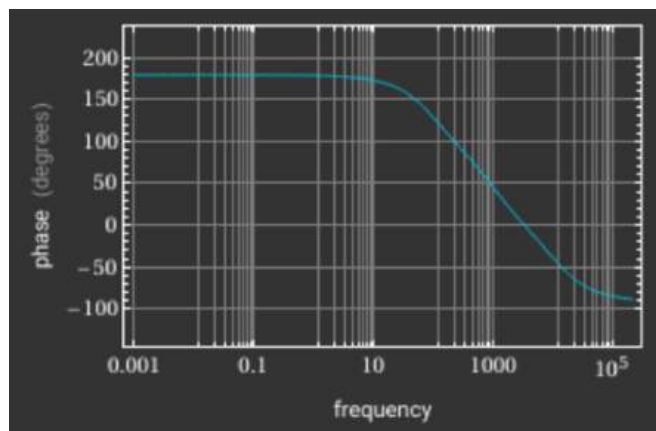
$z_1, z_2 \quad p_1 \quad p_2 \quad p_3$



f) the j in numerator got squared giving the -1 out front

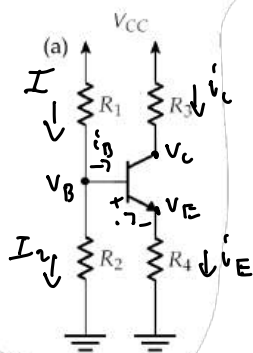


e)



6. For the following circuits, find the base, emitter, and collector voltages and currents. Use

$R_1 = 4.7 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $R_3 = 1 \text{ k}\Omega$, $R_4 = 500 \Omega$, $R_5 = 10 \text{ k}\Omega$, $R_6 = 100 \text{ k}\Omega$, $R_7 = 2.2 \text{ k}\Omega$, $R_8 = 2.7 \text{ k}\Omega$, $C_1 = 1 \mu\text{F}$, $I_1 = 10 \text{ mA}$, $V_{CC} = +12 \text{ V}$, $-V_{EE} = -12 \text{ V}$, and $\beta = 300$.



$$\frac{V_{CC} - V_B}{R_1} = \frac{V_B}{R_2} + \frac{V_B - 0.7}{R_4(\beta + 1)} \Rightarrow V_B = \left(\frac{V_{CC}}{R_1} + \frac{0.7}{R_4(\beta + 1)} \right) \left(\frac{1}{R_2} + \frac{1}{R_4(\beta + 1)} + \frac{1}{R_1} \right)^{-1}$$

$$V_B = 2.098 \text{ [V]}$$

$$V_E = V_B - 0.7 = 1.398 \text{ [V]}$$

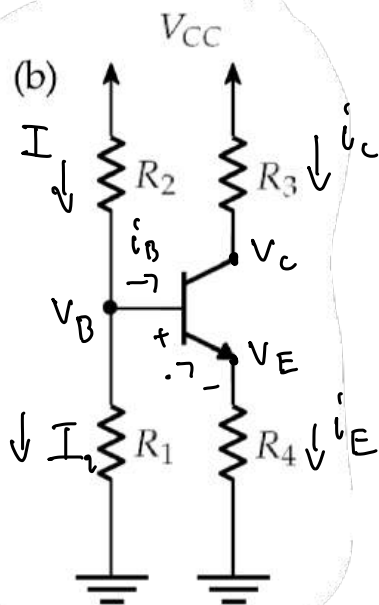
$$I_E = \frac{V_E}{R_4} = \frac{V_B - 0.7}{R_4} = 2.795 \text{ [mA]}$$

$$I_B = \frac{I_E}{\beta + 1} = \frac{V_B - 0.7}{R_4(\beta + 1)} = 9.284 \text{ [}\mu\text{A]}$$

$$I_C = \beta I_B = 2.785 \text{ [mA]}$$

$$V_{CC} - V_C = I_C R_3 \Rightarrow V_C = V_{CC} - I_C R_3$$

$$V_C = 9.214 \text{ [V]}$$



$$\frac{V_{CC} - V_B}{R_2} = \frac{V_B - 0.7}{R_4(\beta + 1)} + \frac{V_B}{R_1}$$

$$V_B = \left(\frac{1}{R_4(\beta + 1)} + \frac{1}{R_1} + \frac{1}{R_2} \right)^{-1} \left(\frac{V_{CC}}{R_2} + \frac{0.7}{R_4(\beta + 1)} \right)$$

$$V_B = 9.845 \text{ [V]}$$

$$I_E = \frac{V_E}{R_4} = 18.29 \text{ mA}$$

$$V_E = V_B - 0.7 = 9.14 \text{ [V]}$$

$$I_B = \frac{I_E}{\beta + 1} = 60.76 \mu\text{A}$$

$$V_C = V_{CC} - I_C R_3 = -6.23 \text{ [V]}$$

$$I_C = \frac{\beta}{\beta + 1} I_E = 18.23 \text{ mA}$$

$$V_C - V_E < 0.2 \text{ V, saturated}$$

$$V_C - V_E = 0.2 \text{ [V]}$$

$$V_E = V_B - 0.7$$

$$I_E = I_C + I_B$$

$$I_E = \frac{V_E}{R_4} = \frac{V_B - 0.7}{R_4}$$

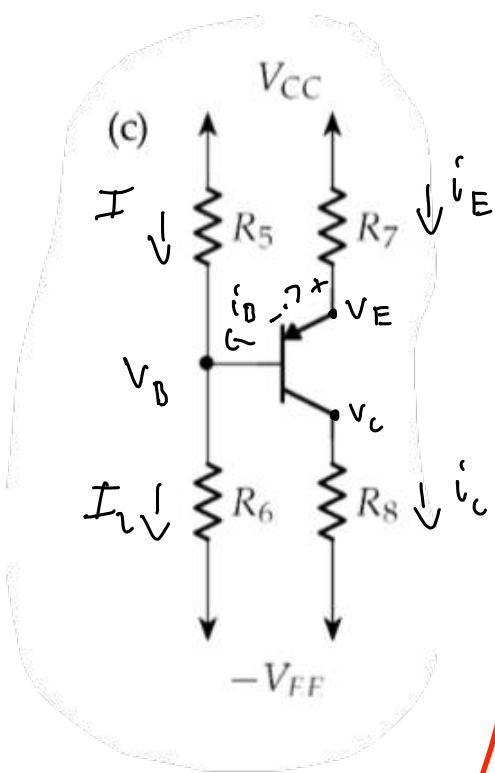
$$I_C = \frac{V_{CC} - V_C}{R_3}$$

$$\frac{V_B - 0.7}{R_4} = \frac{V_{CC} - (0.2 + (V_B - 0.7))}{R_3} + \left(\frac{V_{CC} - V_B}{R_2} - \frac{V_B}{R_1} \right)$$

$$V_B \left(\frac{1}{R_4} + \frac{1}{R_3} + \frac{1}{R_2} + \frac{1}{R_1} \right) = V_{CC} \left(\frac{1}{R_3} + \frac{1}{R_2} \right) + \frac{0.7}{R_4} + \frac{0.5}{R_3}$$

$$V_B = \left(\frac{1}{R_4} + \frac{1}{R_3} + \frac{1}{R_2} + \frac{1}{R_1} \right)^{-1} V_{CC} \left(\frac{1}{R_3} + \frac{1}{R_2} \right) + \frac{0.7}{R_4} + \frac{0.5}{R_3}$$

$$V_B = 6.148 \text{ [V]} \quad V_E = 5.448 \quad V_C = 5.648 \quad I_E = 10.9 \mu\text{A} \quad I_C = 6.35 \text{ mA} \quad I_B = 4.54 \mu\text{A}$$



$$\frac{V_{CC} - V_B}{R_5} + i_B = \frac{V_B - V_{EE}}{R_4}$$

$$i_B = \frac{i_E}{\beta + 1} = \frac{V_{CC} - V_E}{R_7(\beta + 1)} = \frac{V_{CC} - (V_B + 0.7)}{R_7(\beta + 1)}$$

$$\frac{V_{CC} - V_B}{R_5} + \frac{V_{CC} - (V_B + 0.7)}{R_7(\beta + 1)} = \frac{V_B - V_{EE}}{R_4}$$

$$V_B \left(\frac{1}{R_4} + \frac{1}{R_7(\beta + 1)} + \frac{1}{R_5} \right) = V_{CC} \left(\frac{1}{R_5} + \frac{1}{R_7(\beta + 1)} \right) - \frac{0.7}{R_7(\beta + 1)} + \frac{V_{EE}}{R_4}$$

$$V_B = 9.84 \text{ [V]}$$

$$i_B = \frac{i_E}{\beta + 1} = 2.21 \text{ mA}$$

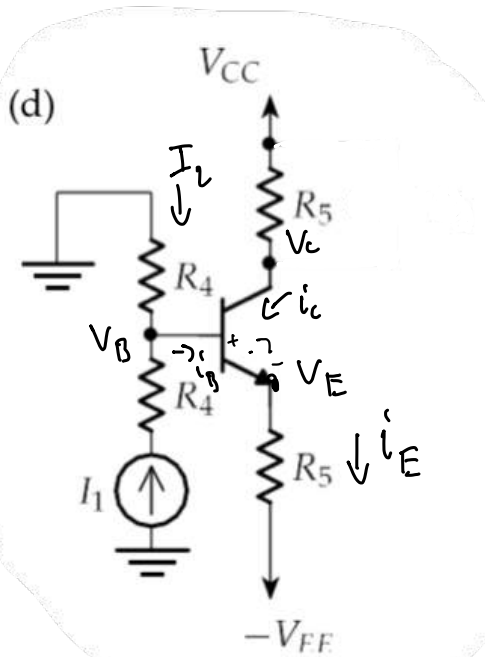
$$V_E = V_B + 0.7 = 10.54 \text{ [V]}$$

$$i_E = \frac{V_{CC} - V_E}{R_7} = 664.4 \text{ mA}$$

$$V_C = V_{EE} + i_C R_8$$

$$i_C = \frac{\beta}{\beta + 1} i_E = 662.23 \text{ mA}$$

$$V_C = -10.21 \text{ [V]}$$



$C_1 = \text{Open in dc}$

$$\frac{-V_B}{R_4} + I_1 = i_B$$

$$i_B = \frac{i_E}{\beta + 1} = \frac{V_E - V_{EE}}{R_5(\beta + 1)} = \frac{(V_B - 0.7) - V_{EE}}{R_5(\beta + 1)}$$

$$\frac{(V_B - 0.7) - V_{EE}}{R_5(\beta + 1)} + \frac{V_B}{R_4} = I_1$$

$$V_B \left(\frac{1}{R_5(\beta + 1)} + \frac{1}{R_4} \right) = I_1 + \frac{V_{EE}}{R_5(\beta + 1)} + \frac{0.7}{R_5(\beta + 1)}$$

$$V_B = \left(\frac{1}{R_5(\beta + 1)} + \frac{1}{R_4} \right)^{-1} \left(I_1 + \frac{V_{EE}}{R_5(\beta + 1)} + \frac{0.7}{R_5(\beta + 1)} \right)$$

$$V_B = 4.997 \text{ [V]}$$

$$V_E = V_B - 0.7 = 4.297 \text{ [V]}$$

$$i_E = \frac{V_E - V_{EE}}{R_5} = 1.63 \text{ mA}$$

$$V_C = V_{CC} - i_C R_5 = -4.24 \text{ [V]}$$

$$i_C = \frac{\beta}{\beta + 1} i_E = 1.62 \text{ mA}$$

$V_C - V_E < 0.2 \therefore \text{Saturated}$

$$i_E = i_C + i_B$$

$$i_E = \frac{V_E - V_{EE}}{R_5} = \frac{V_B - 0.7 - V_{EE}}{R_5}$$

$$i_B = \frac{-V_B}{R_4} + I_1$$

$$i_C = \frac{V_{CC} - V_C}{R_5} = \frac{V_{CC} - (I_2 + V_E)}{R_5} = \frac{V_{CC} - (I_2 + (V_B - 0.7))}{R_5}$$

$$\frac{V_B - 0.7 - V_{EE}}{R_5} = \frac{V_{CC} - (I_2 + (V_B - 0.7))}{R_5} + \frac{-V_B}{R_4} + I_1$$

$$V_B \left(\frac{1}{R_5} + \frac{1}{R_5} + \frac{1}{R_4} \right) = \frac{V_{CC}}{R_5} + \frac{I_2}{R_5} + \frac{0.7}{R_5} + \frac{V_{EE}}{R_5} + I_1$$

$$V_B = 4.6 \text{ [V]} \quad V_E = 3.9 \text{ [V]} \quad V_C = 4.7 \text{ [V]}$$

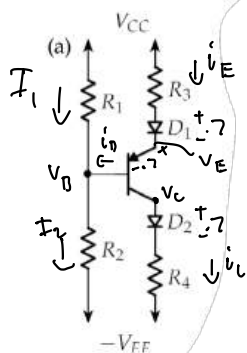
$$i_B = 0.782 \text{ mA}$$

$$i_E = 1.59 \text{ mA}$$

$$i_C = 0.799 \text{ mA}$$

7. For the following circuits, find the base, emitter, and collector voltages and currents. Use

$R_1 = 1 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $R_3 = 3.3 \text{ k}\Omega$, $R_4 = 5 \text{ k}\Omega$, $R_5 = 10 \text{ k}\Omega$, $R_6 = 1 \text{ k}\Omega$, $I_1 = 1 \text{ mA}$, $V_{CC} = +10 \text{ V}$, $-V_{EE} = -10 \text{ V}$, and $\beta = 150$. Use a constant voltage drop model for the diodes with a forward bias potential of 0.7 V .



$$V_{CC} - V_B = i_E R_3 + 0.7 + 0.7$$

$$V_B = V_{CC} - i_E R_3 - 0.7 - 0.7$$

$$V_B = 8.6 - i_E R_3$$

$$V_B = 8.6 - (\beta + 1) i_B R_3$$

$$V_B = 8.8 \text{ (mV)}$$

$$V_E = V_B + 0.7 = 0.709 \text{ (V)}$$

$$V_C - V_{EE} = 0.7 + i_C R_4$$

$$V_C = V_{EE} + 0.7 + i_C R_4$$

$$V_C = 3.631 \text{ (V)}$$

$$i_B = \frac{V_B - V_{EE}}{R_2} = \frac{V_{CC} - V_B}{R_1}$$

$$(i_B = V_B \left(\frac{1}{R_2} + \frac{1}{R_1} \right) - \frac{V_{CC}}{R_1} - \frac{V_{EE}}{R_2}) 1 \text{ k}$$

$$1 \text{ k} i_B = 2 V_B - 10 - (-10)$$

$$1 \text{ k} i_B = 2(8.6 - 498300 i_B)$$

$$1 \text{ k} i_B + 996600 i_B = 17.2$$

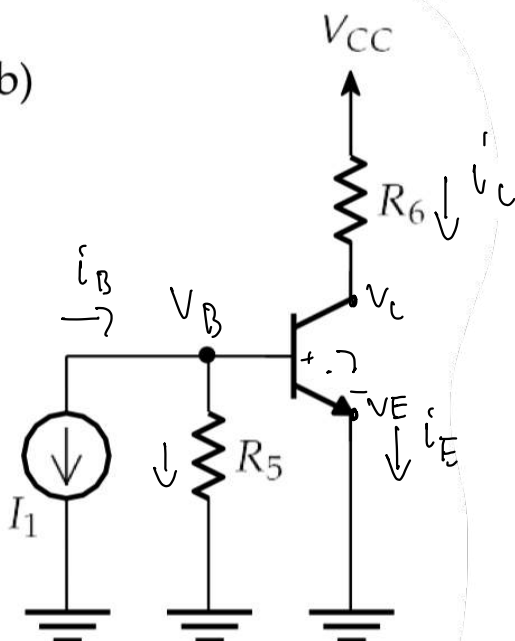
$$997600 i_B = 17.2$$

$$i_B = 17.24 \text{ (}\mu\text{A)}$$

$$i_E = (\beta + 1) i_B = 2.403 \text{ (mA)}$$

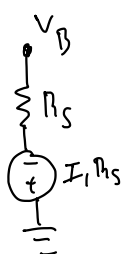
$$i_C = \beta i_B = 2.586 \text{ (mA)}$$

(b)



$$V_E = 0$$

$$V_B = V_E + 0.7 = 0.7 \text{ (V)}$$



$$V_B + I_1 R_5 = 0$$

$$V_B = -I_1 R_5$$

$$V_B = -10 \text{ (V)}$$

$$V_E = 0$$

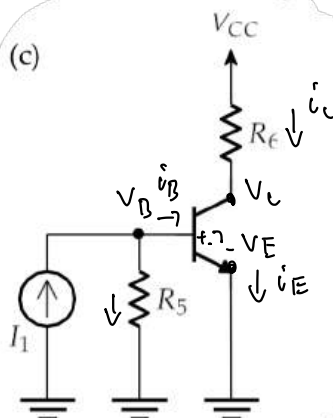
$$V_C = V_{CC} - i_C R_6 \Rightarrow V_C = 10 \text{ (V)}$$

$$i_B = -I_1 - \frac{V_B}{R_5}$$

$$i_B = -1.07 \text{ (mA)} < 0 \therefore \text{cutoff}$$

$$\text{so } i_B = i_C = i_E = 0$$

(c)



$$V_E = 0$$

$$V_B = V_E + 0.7 = 0.7 \text{ (V)}$$

$$V_C = V_{CC} - R_6 i_C = -129.5 \text{ (V)}$$

$$V_C - V_E < 0 \therefore \text{saturated}$$

$$i_E = i_C + i_B$$

$$i_E = \frac{V_{CC} - V_C}{R_6} + I_1 - \frac{V_B}{R_5}$$

$$i_E = 10.73 \text{ (mA)}$$

$$i_C = \frac{V_{CC} - V_C}{R_6} = 9.8 \text{ (mA)}$$

$$i_B = 0.93 \text{ (mA)}$$

$$V_C - V_E = 0.2$$

$$V_B - V_E = 0.7$$

$$V_E = 0$$

$$V_C = 0.2$$

$$V_B = 0.7$$

$$-I_1 + \frac{V_B}{R_5} + i_B = 0$$

$$i_B = I_1 - \frac{V_B}{R_5}$$

$$i_B = 930 \text{ (}\mu\text{A)}$$

$$i_E = i_B (\beta + 1)$$

$$i_C = \beta i_B$$

$$i_E = 140.43 \text{ (mA)}$$

$$i_C = 139.5 \text{ (mA)}$$